# Math for CS & AI: Homework 9

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(Adapted from earlier versions of this course taught by Andrew Yao)

**Due Date: May 23, 2026**

## Study Tips

Some students have asked for additional exercises to practice their problem-solving skills. These exercises are not required, but if you are interested, you may find some useful problems in the following textbook:

- *All of Statistics: A Concise Course in Statistical Inference*, by Larry Wasserman
  - Chapter 2: Random Variables
  - Chapter 3: Expectation

You may also want to practice your calculus skills to truly master continuous probability. Check any standard calculus textbook for exercises on integration and differentiation.

**Note:** Skip the problems that require more knowledge than what is covered in this course.

## Problem Set

- Exercises 4.18–4.19 from Chapter 4 and 5.1–5.8 from Chapter 5 of the lecture notes: https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

## Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

# Additional Hints

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.




Think twice before scrolling down!

# Hints

## Problem 4.18

- Read the lecture notes on a first look at the CLT.
- You only need to change the argument there slightly.

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## Problem 4.19

- For the CGF, wow the probability mass function of the binomial distribution is so complicated...
- Write it as a sum of Bernoulli random variables.
- Use the CGFs of Bernoulli random variables to compose the CGF of the binomial random variable.
- For the second formula, write down the probability mass function of the binomial distribution first.
- There will be a binomial coefficient. Expand it by definition.

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## Problem 5.1

- If you do not know how to proceed, go through the examples on countable and uncountable sets in the lecture notes!
- Basically, if a set can be listed as a sequence, then after an injection, it is a sequence of elements in the target set.
- Be careful: some elements are repeated in the sequence. You should skip the repeated elements.

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## Problem 5.2

- If you do not know how to proceed, go through the examples on countable and uncountable sets in the lecture notes!
- Use the conclusion from the previous problem will make your life easier.

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## Problem 5.3

- What is the Taylor series of $e^{\lambda x}$?
- For (c), you can either use brute force by computing the convolution by hand, or read the lecture notes on Gaussian linearity for inspiration.

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## Problem 5.4

- (a)(c) Find the antiderivative of $e^{-\lambda x}$ first.
- (b) Having a tough time on finding the antiderivative of $xe^{-\lambda x}$? Try integration by parts.
- (d) Read the lecture notes on Gaussian linearity for inspiration.

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## Problem 5.5

- No hints here. This is just a simple problem to get you familiar with the concepts of CDFs and quantile functions.

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## Problem 5.6

- It may help your understanding if you draw some sketches of the CDF and the quantile function first.
- (a) There is a useful theorem in the lecture notes.
- (b) Just need to show that $\Pr[F_{\mathcal{D}}^{-1}(Z) \leq a] = \Pr[X \leq a]$ for $X \sim \mathcal{D}$.
- $F_{\mathcal{D}}$ is non-decreasing: $F_{\mathcal{D}}^{-1}(Z) \leq a \iff F_{\mathcal{D}}(F_{\mathcal{D}}^{-1}(Z)) \leq F_{\mathcal{D}}(a) \overset{?}{\iff} Z \leq F_{\mathcal{D}}(a)$.
- If $Z > F_{\mathcal{D}}(a)$, then? If $Z \leq F_{\mathcal{D}}(a)$, then?

## Problem 5.7

- Moment generating function.

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## Problem 5.8

- Integration by parts.